WiFi

SSID: UPV-PUBLIC
PSK:

Before we start!

1. Download Arduino AppLab
   https://bit.ly/arduino-applab
2. Sign-up to Edge Impulse
   https://edgeimp.com/roscon-spain
3. And sign-up to Arduino with **the same e-mail**
   https://bit.ly/arduino-account

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ARDUINO

EDGE IMPULSE
a Qualcomm company

Creando robots con ROS2 y Arduino VENTUNO Q

Marc Pous Marín - Edge Impulse DevRel
Julián Caro Linares - Arduino Tech Evangelist



ARDUINO

EDGE IMPULSE
a Qualcomm company

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github.com/jcarolinares/

Marc Pous

@gy4nt
linkedin.com/in/marcpous

github.com/mpous/




ARDUINO

¿Quién ha usado Arduino?



¿Quién usa MicroROS?

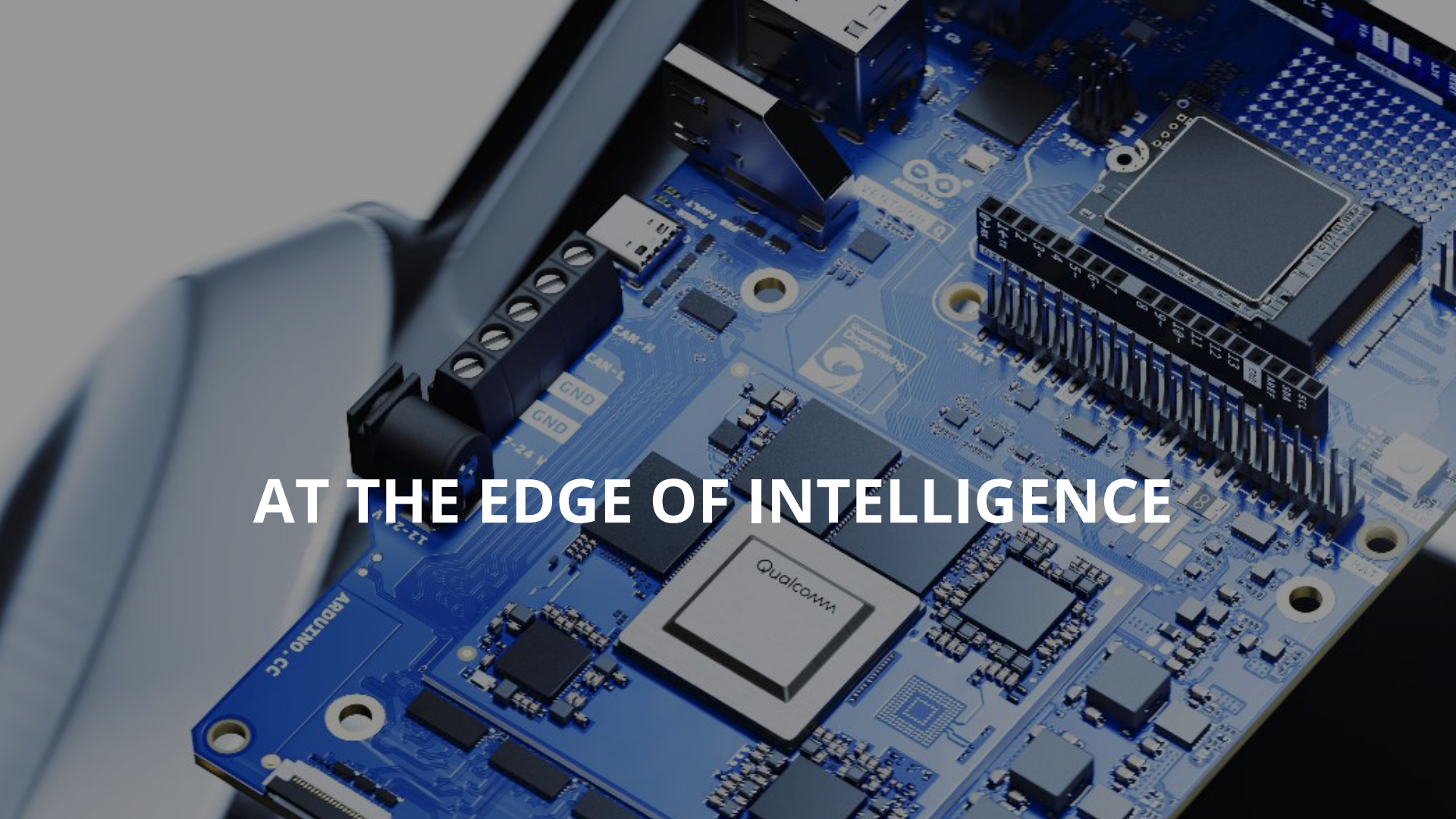


¿Quién usa Arduino con ROS?



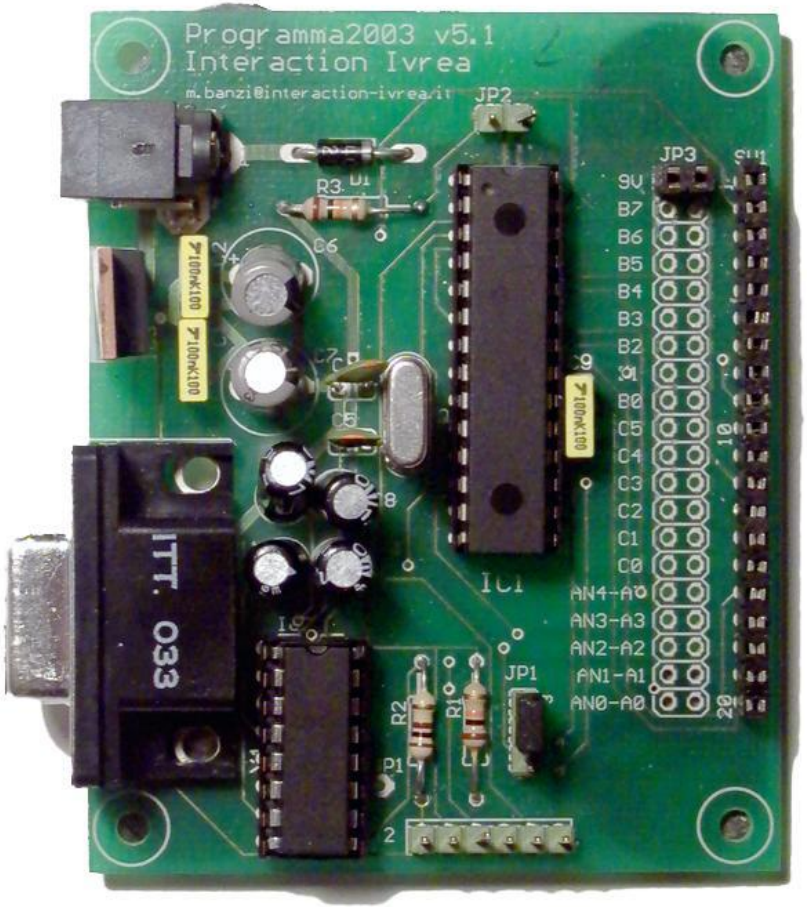
¿Quién conoce
a Naranjito?

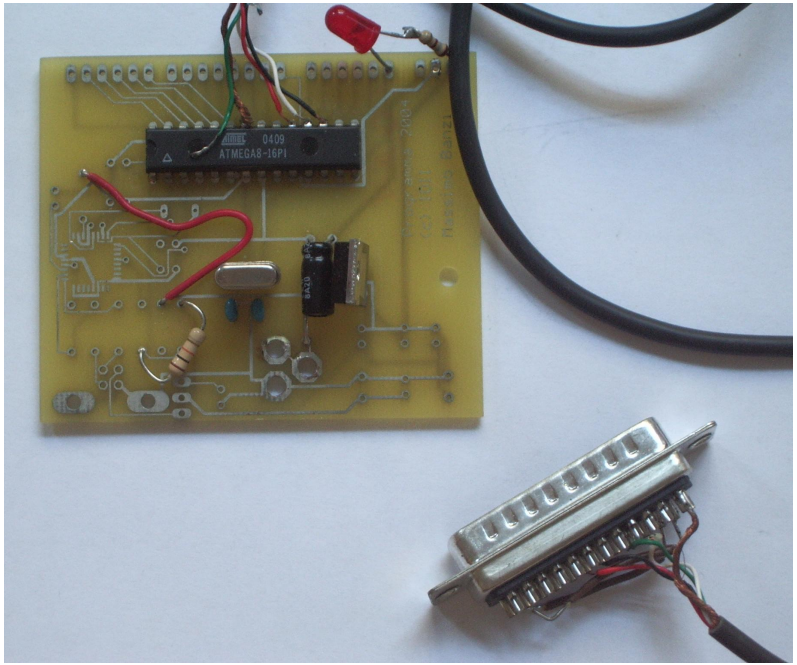



AT THE EDGE OF INTELLIGENCE
Qualcomm
CAN-H
CAN-L
GND
GND
ARDUINO.CC


Where it all started:
Interaction Design Institute
Ivrea, Italy (2001)

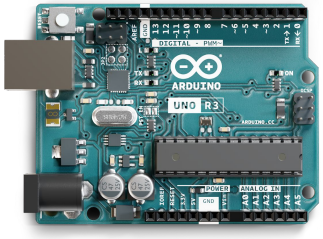



Programma2003 v5.1
Interaction Ivrea
m.banzi@interaction-ivrea.it
JP2
JP3
SV1
R3
D1
C6
C7
C5
100nK100
100nK100
100nK100
9V
B7
B6
B5
B4
B3
B2
B1
B0
C5
C4
C3
C2
C1
C0
AN4-A4
AN3-A3
AN2-A2
AN1-A1
AN0-A0
IC1
IC2
JP1
R2
R1
P1
X4
ITT 033


ATMEGA8-16PI
0409

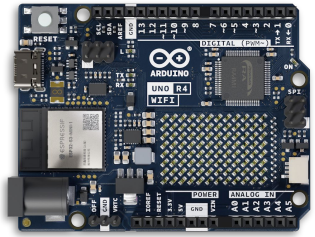


Arduino (2005)



UNO R3

8-Bit MCU



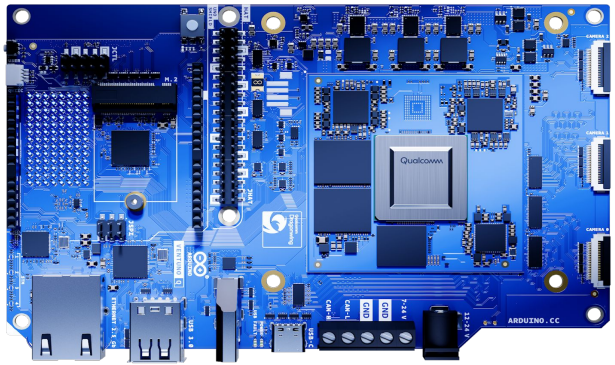
UNO R4

32-Bit MCU



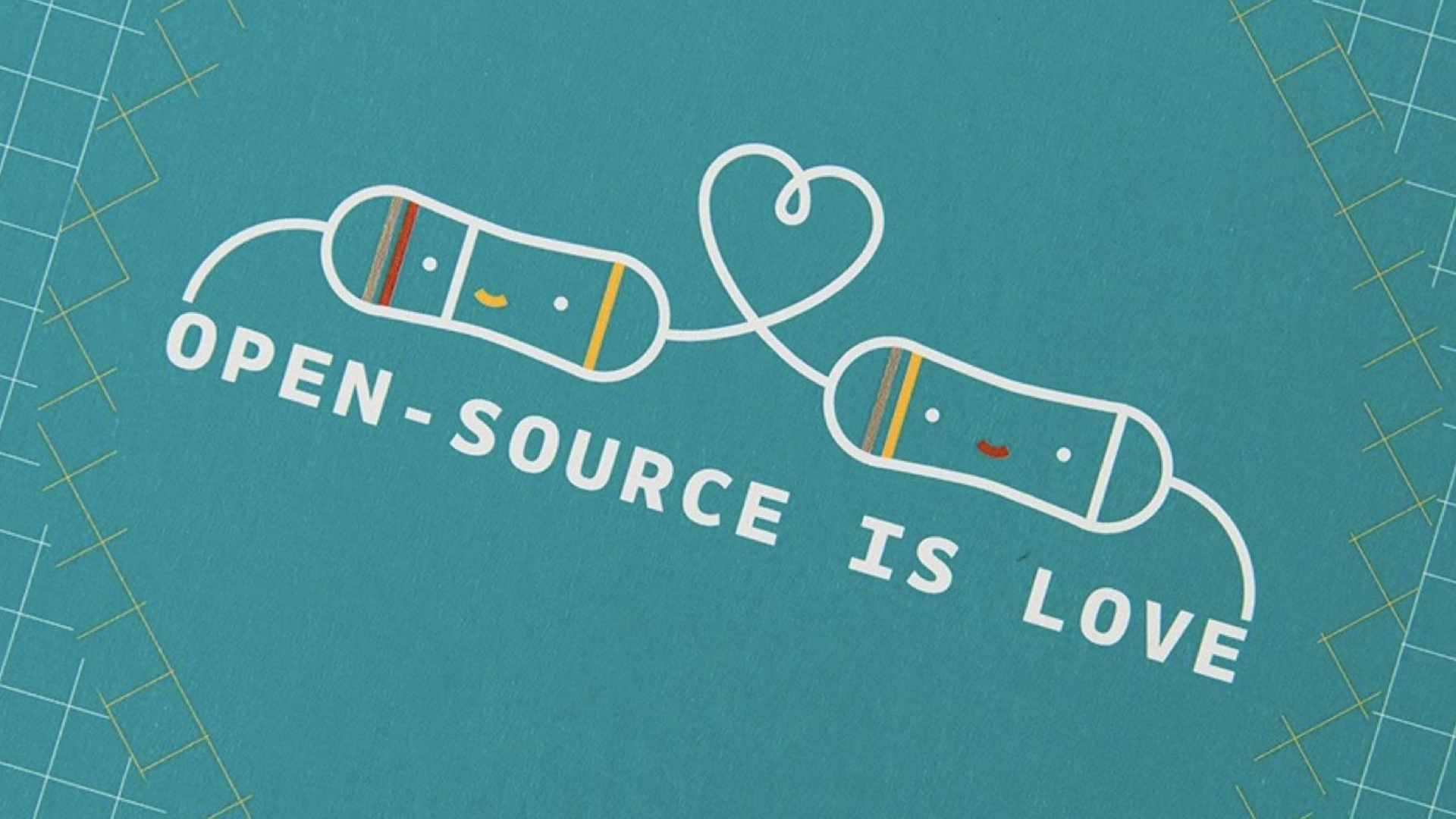
UNO Q

CPU + MCU



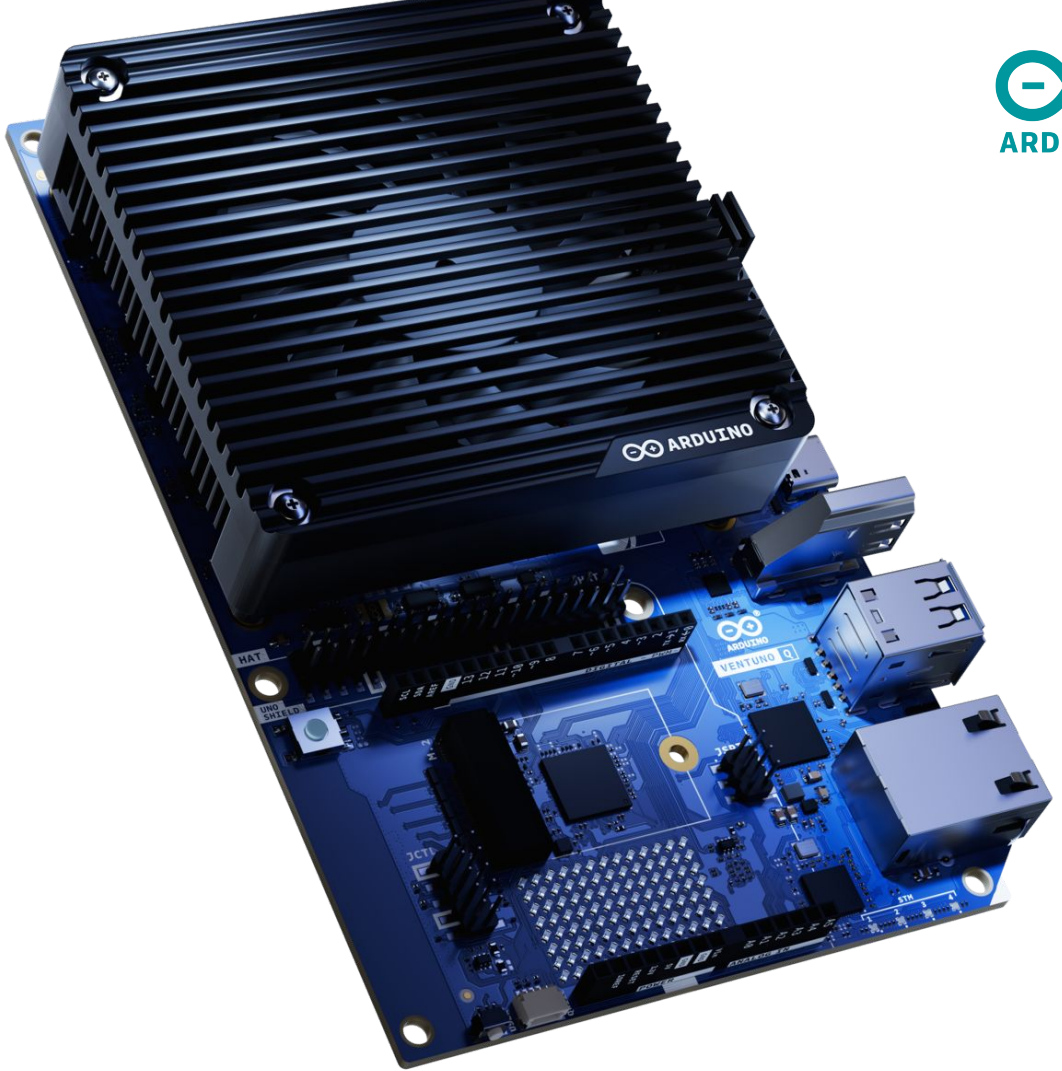
VENTUNO Q

CPU + NPU + MCU


OPEN-SOURCE IS LOVE

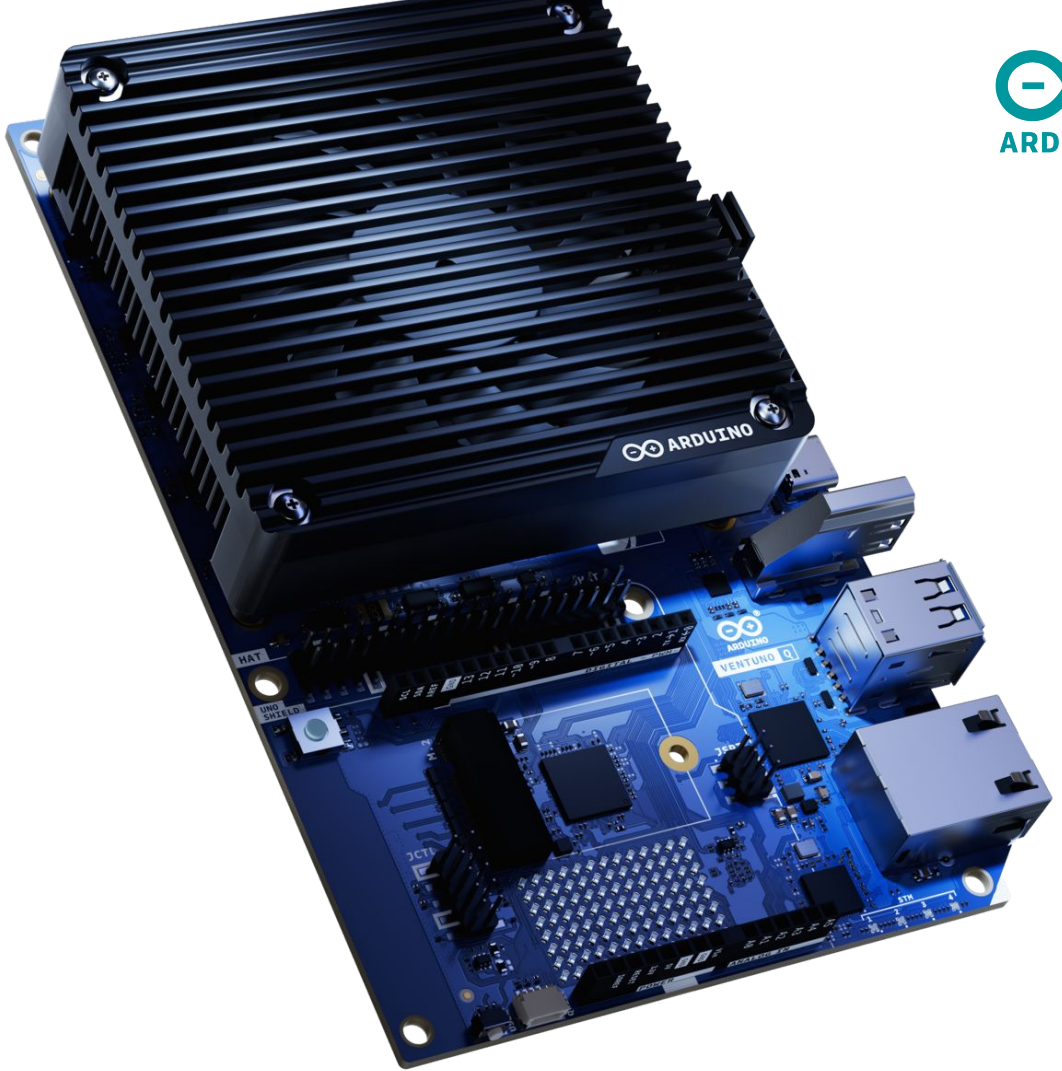
Created for Physical AI

Arduino VENTUNO Q

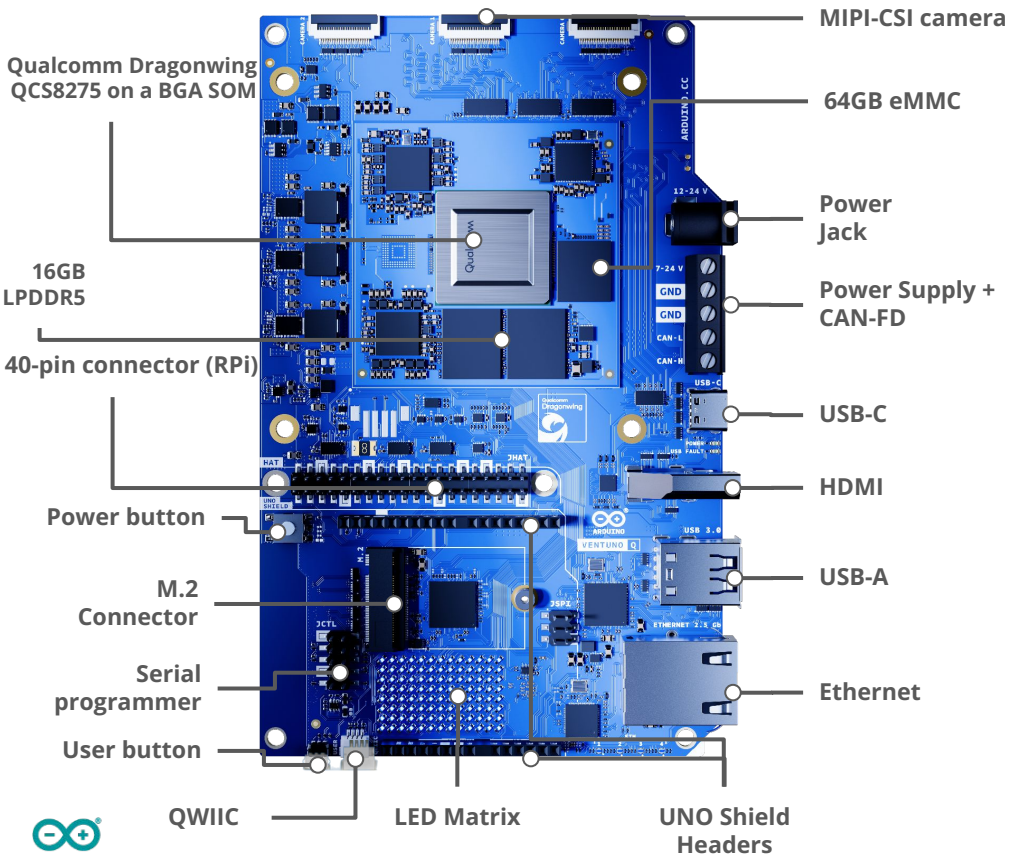


Created for ~~Physical AI~~
Robotics

Arduino VENTUNO Q



Tech Specs



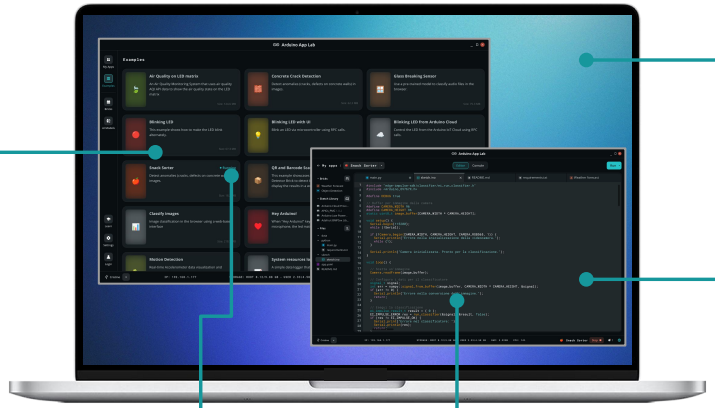
TOP VIEW



BOTTOM VIEW

Dual-Brain Architecture: Linux Meets Arduino

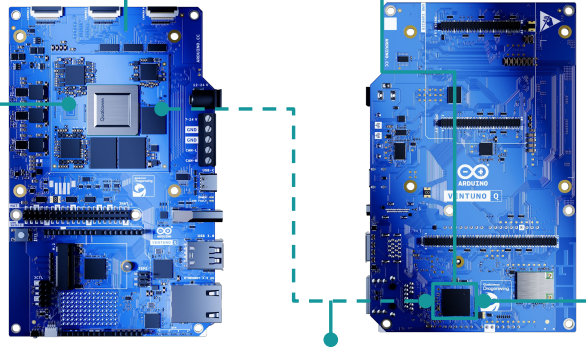
Arduino App Lab combines **Arduino sketches, Python** scripts, and **AI models** into ready-to-run apps



Linux Ubuntu OS / Ustream Debian OS (coming soon)

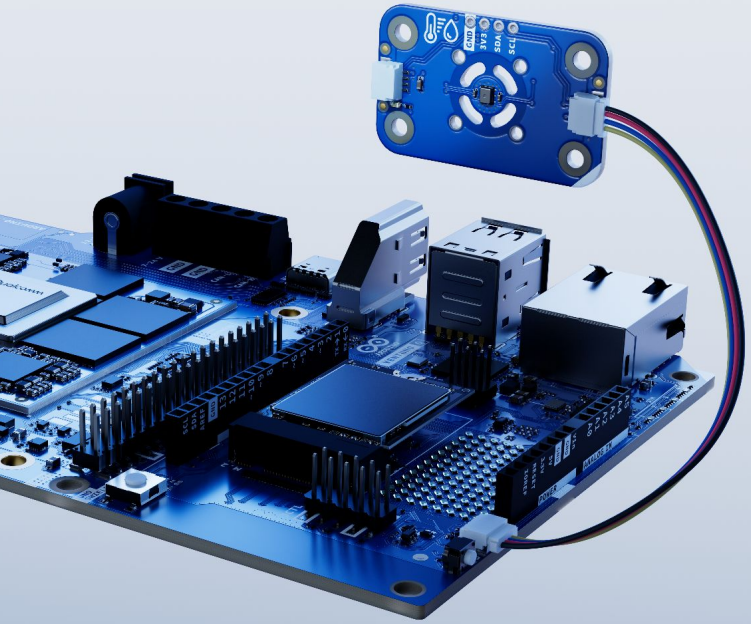
Arduino Core and **Sketch** experience running on **Zephyr OS**

Qualcomm Dragonwing ™ IQ-8275 (up to 40 Dense TOPS)



STM32H5 for low-latency actuation and motor control

Remote Procedure Call (RPC)



Shields, Carriers and Modulino nodes: Plug Into Possibilities

Unlock seamless innovation with the VENTUNO Q. It merges the vast **Arduino UNO shield** and **Raspberry Pi® Hat** ecosystem with a Qwiic connector for instant access to **Modulino® nodes** and sensors, no soldering required.

Designed for versatility, the high-speed headers tap into a dual-brain architecture to bridge **UNO and VENTUNO carrier** compatibility, enabling complex applications requiring high-speed cameras, displays, audio or precise motor control.

No soldering needed

Focus on building your next idea, little wiring and no soldering will help you to prototype faster.

Polarized connector

No worries about wrong connections, there is only a way to connect the board and the modules.

Chain-able MODULINOS

Connect multiple Qwiic compatible modules one after the other and expand the possibilities of your next project.

Tech Specs

Hardware

Microprocessor (MPU)	• Qualcomm Dragonwing™ IQ8 (QCS8275): ○ Octa-core Kryo Gen 6 CPU ○ Adreno 623 GPU ○ Hexagon Tensor AI Processor (NPU): up to 40 dense TOPS ○ Qualcomm Spectra 690 ISP	**Microcontroller (MCU)**	• STM32H5F5: ○ Arm® Cortex® M33 at 250MHz ○ 4MB flash ○ 1.5MB RAM
RAM	• 2x8GB LPDDR5	**Power Supply**	• From USB-C connector 9-20 VDC max at 3 A • 5.5x2.1 mm Power Jack 7-24 VDC • Screw Terminal 7-24 VDC • 7-24 VDC on JOMEGA
Storage	• 64GB eMMC • M.2 connector for NVME Gen.4 external storage	**USB**	• 1× USB-C port with host/device role switching, power role switch and video output • 2x USB 3.0 Type A • 2x USB 3.0 on JOMEGA header
Connectivity	• Wi-Fi® 6 2.4/5/6 GHz with onboard antenna • Bluetooth® 5.3 with onboard antenna • 1x 2.5Gbit RJ45 Ethernet	**CAN**	• 1x CAN-FD PHY on screw terminal • 3x CAN-FD (no PHY) on JOMEGA header • 1x CAN-FD (no PHY) on UNO Shield headers
Camera	• USB camera support • 3x MIPI CSI connectors muxed with 2x MIPI CSI on JMEDIA header	**Interfaces**	I2C/I3C PSSI SPI GPIO PWM JTAG UART ADC
Video	• 1x HDMI muxed with MIPI DSI on JMEDIA header • Video output (DP Alt mode) support via USB-C • MIPI DSI pins on JMEDIA header	**Extra**	• 4× RGB user-controllable LEDs • 8x13 Blue LED Matrix • 1x Qwiic connector voltage 3V3, I2C • 1x User push-button • 1x reset button • JCTL: MPU Remote Debug connector • JTAG port on JOMEGA
Audio	• 2x Microphone IN / Headphone OUT / Ear OUT / Line OUT on JMISC header and JOMEGA	**Dimensions**	• 160x100x25.8 mm
Temperature Range	• Commercial Temperature Range: -10 °C to +60 °C (14 °F to 140 °F)		

Tech Specs

Software

MPU Operating System	• Linux Canonical UBUNTU with 10-years Ubuntu PRO License • Linux Debian OS with upstream support (coming soon)	**Containerization**	• Docker and Docker Compose support
Real-time Operating System	• Arduino Core on Zephyr OS	**Supported Operating Systems for Arduino App Lab**	Windows: Windows 10 or later (64-bit) macOS: macOS 11 or later (64-bit) Linux: Ubuntu 22.04 or later, and Debian Trixie (64-bit)

ubuntu

debian

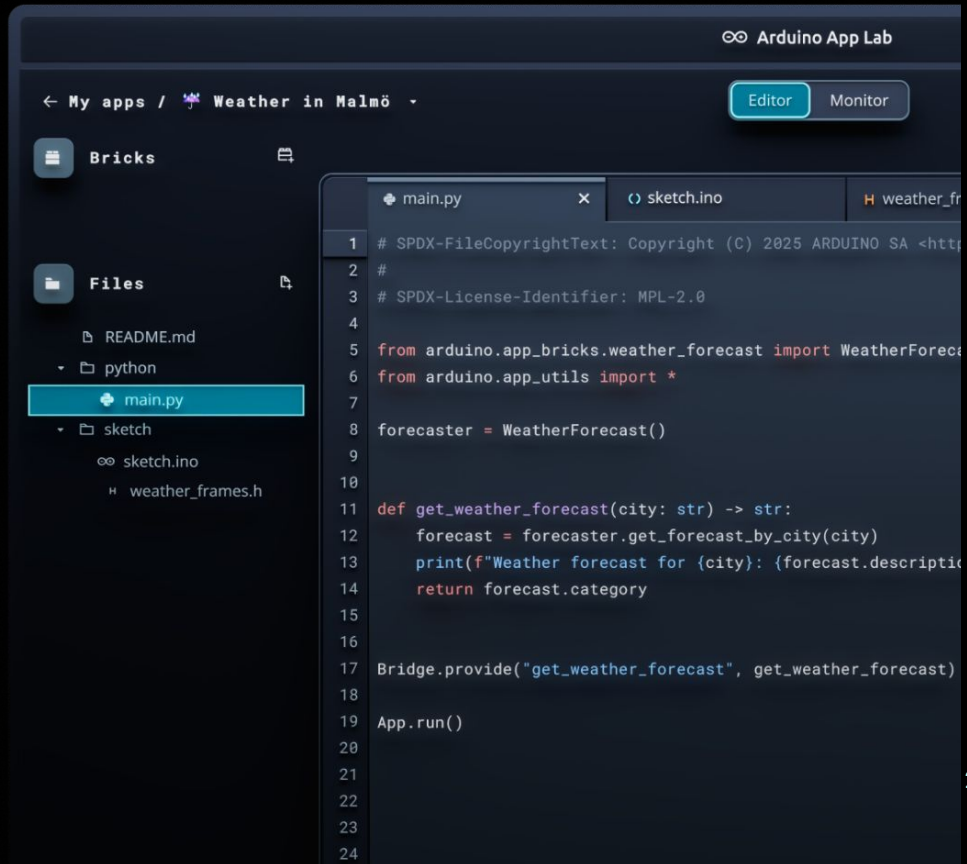
ARDUINO

Zephyr®

docker

Arduino ® App Lab: A Unified Development Experience that Removes Barriers

The Arduino App Lab is the converging point of development across **NPU, CPU, GPU,** and **MCU** platforms, uniting embedded (sketch) programming, Linux development, and edge AI.
https://bit.ly/arduino-applab



Two Setups
One Experience

Standalone Mode

App Lab runs directly on VENTUNO Q, just plug in a monitor, keyboard, and mouse for an all-in-one development environment. No PC needed.



PC-Connected Mode

Connect the VENTUNO Q to your computer (via USB-C or via network mode) and run App Lab on your PC for a smooth and familiar development experience.



Powerful AI Models, running at the Edge



Generative AI
Local LLMs and multimodal models for on-device reasoning, vision-language tasks, and image-to-text generation.



Audio Intelligence
Edge-optimized keyword spotting, speech-to-text, text-to-speech, and acoustic event recognition.



Vision AI
Real-time object detection, classification, segmentation, gesture/pose estimation, tracking, and depth-aware perception.



Robotics
Perception models for navigation, sensor-fusion pipelines, action-triggering logic, and much more

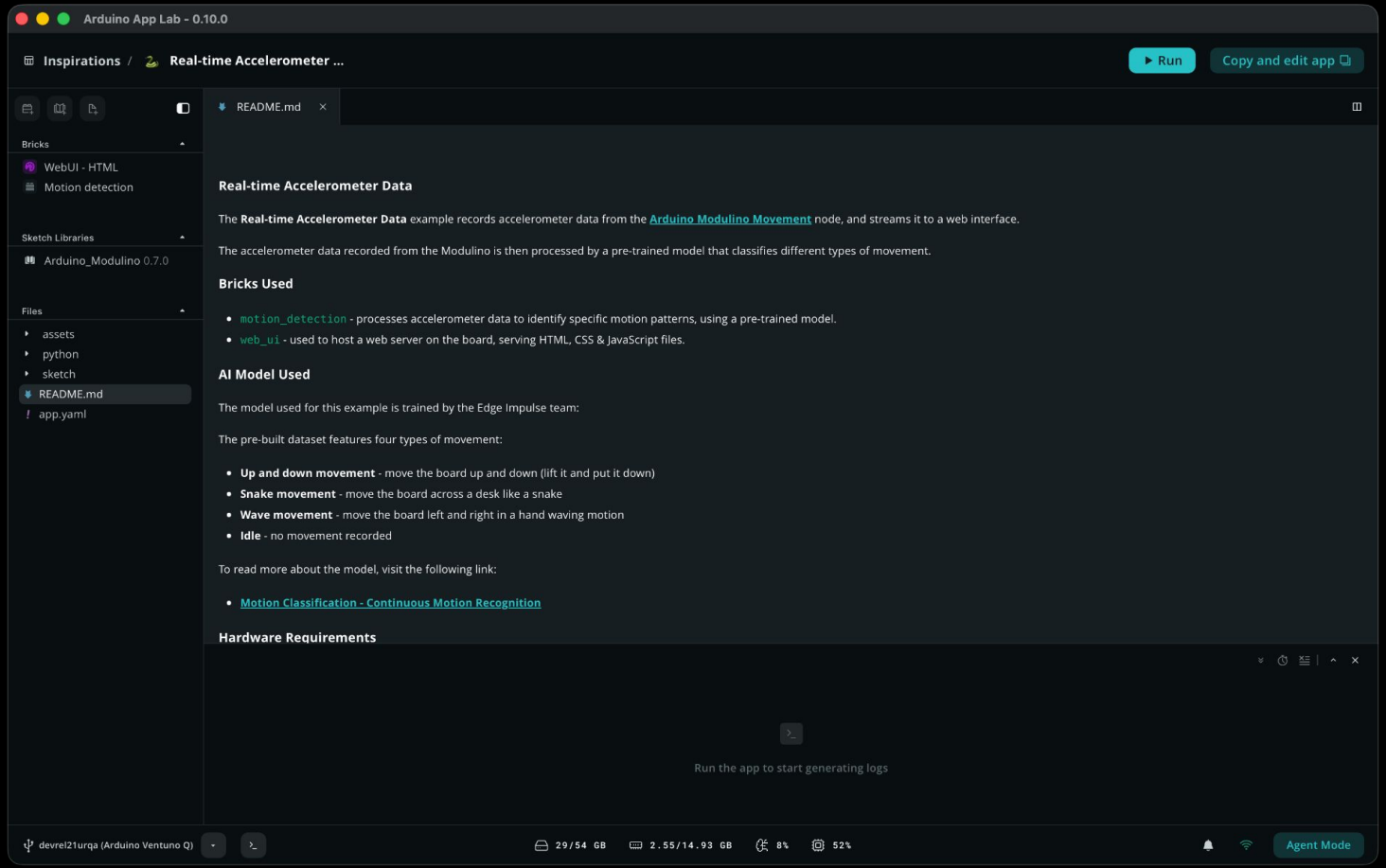




EDGE IMPULSE



Let’s try Arduino Applab


Arduino App Lab - 0.10.0
Inspirations / Real-time Accelerometer ...
Run
Copy and edit app
README.md
Bricks
WebUI - HTML
Motion detection
Sketch Libraries
Arduino_Modulino 0.7.0
Files
assets
python
sketch
README.md
app.yaml
Real-time Accelerometer Data
The Real-time Accelerometer Data example records accelerometer data from the Arduino Modulino Movement node, and streams it to a web interface.
The accelerometer data recorded from the Modulino is then processed by a pre-trained model that classifies different types of movement.
Bricks Used
motion_detection - processes accelerometer data to identify specific motion patterns, using a pre-trained model.
web_ui - used to host a web server on the board, serving HTML, CSS & JavaScript files.
AI Model Used
The model used for this example is trained by the Edge Impulse team:
The pre-built dataset features four types of movement:
Up and down movement - move the board up and down (lift it and put it down)
Snake movement - move the board across a desk like a snake
Wave movement - move the board left and right in a hand waving motion
Idle - no movement recorded
To read more about the model, visit the following link:
Motion Classification - Continuous Motion Recognition
Hardware Requirements
Run the app to start generating logs
devrel21urqa (Arduino Ventuno Q)
29/54 GB
2.55/14.93 GB
8%
52%
Agent Mode

arduino-cli
applab-cli



Arduino App Lab - 0.10.0

Inspirations / Detect Objects on Camera

Run | Copy and edit app

README.md | Video Object Detection

Bricks
- Video Object Detection
- WebUI - HTML

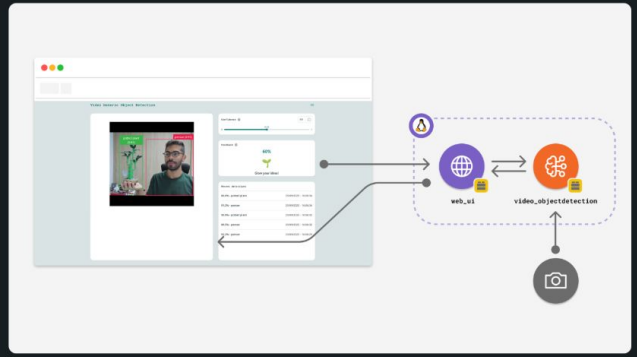
Sketch Libraries
No sketch libraries added yet

Files
- assets
- python
- README.md
- app.yaml

Detect Objects on Camera

The **Detect Objects on Camera** example lets you detect objects on a live feed from a USB camera and visualize bounding boxes around the detections in real-time.

Note: This example must be run in **Network Mode** in the Arduino App Lab, since it requires a USB-C hub and a USB camera.



This example uses a pre-trained model to detect objects on a live video feed from a camera. The workflow involves continuously getting the frames from a USB camera, processing it through an AI model using the

Run the app to start generating logs

devrel21urqa (Arduino Ventuno Q) | 29/54 GB | 2.44/14.93 GB | 9% | 55% | Agent Mode

AppLab → Detect Objects on Camera

Custom ML model ?



EDGE IMPULSE
a Qualcomm company

Train your Custom ML model with Edge Impulse

Copyright © Edge Impulse Inc.

What is Edge Impulse?

The End-to-End ML Ops platform for model development & model deployment to the edge:

Ingest & Build Data Sets

- Supports sensor, audio, and image data
- Collect and visualize datasets from real-world environments

Iterate and Develop Models

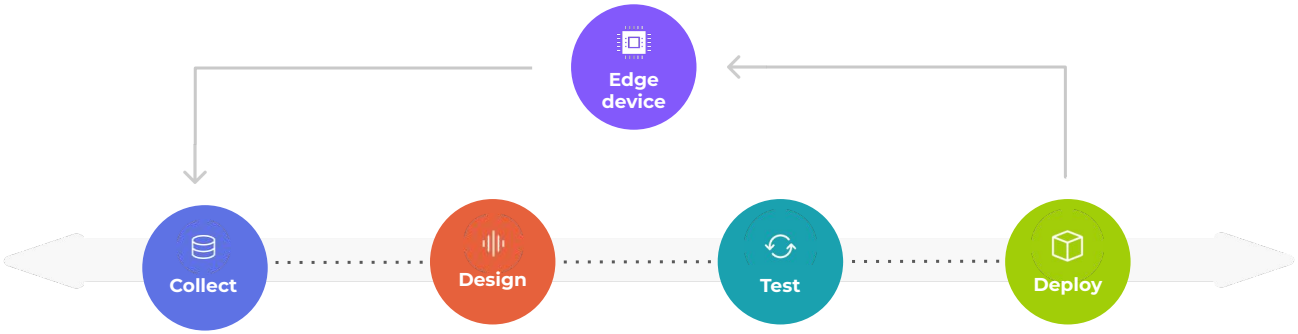
- Extract features and design model architectures
- Rapid iteration with built-in tools for labeling, training, and tuning

Optimize & Deploy

- Profile model performance for target hardware
- Compress, quantize, and export to MCUs, NPUs, and edge devices

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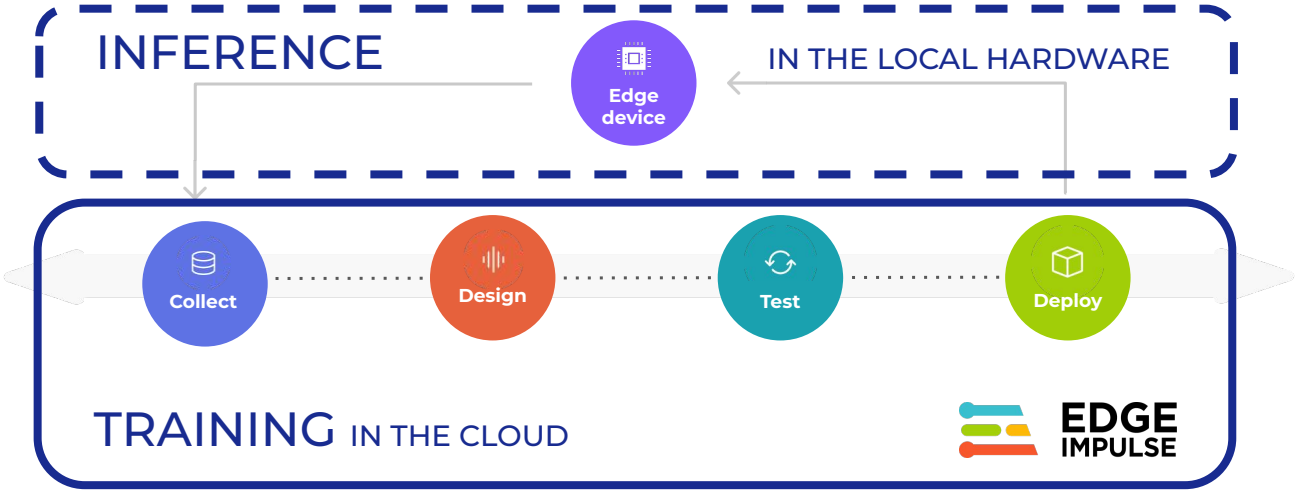
Edge AI



EDGE IMPULSE

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Edge AI

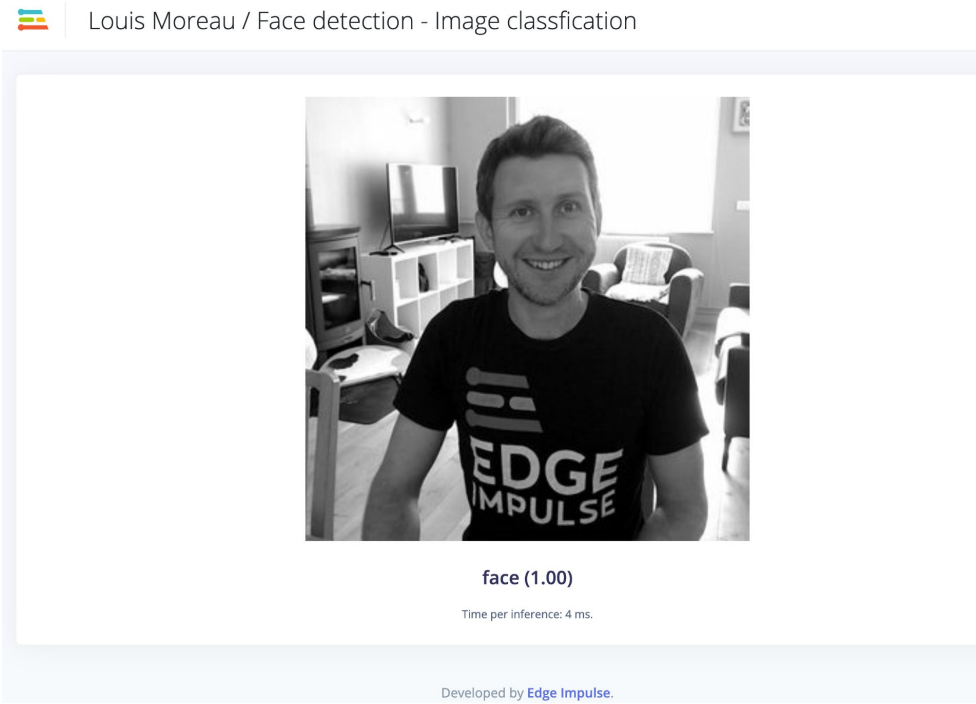


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Image classification

The question the model is trying to answer is:

“Is there a face or not in the image?”

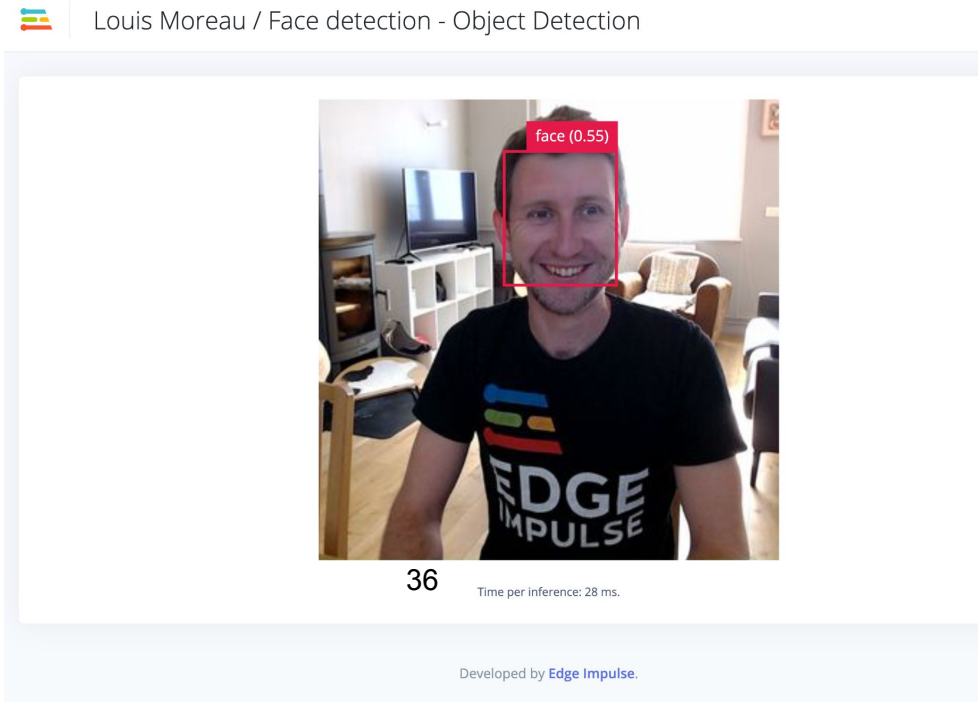


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Object Detection using Bounding Boxes

The question the model is trying to answer is:

“Are there faces in the image, where **and what size are they?”**

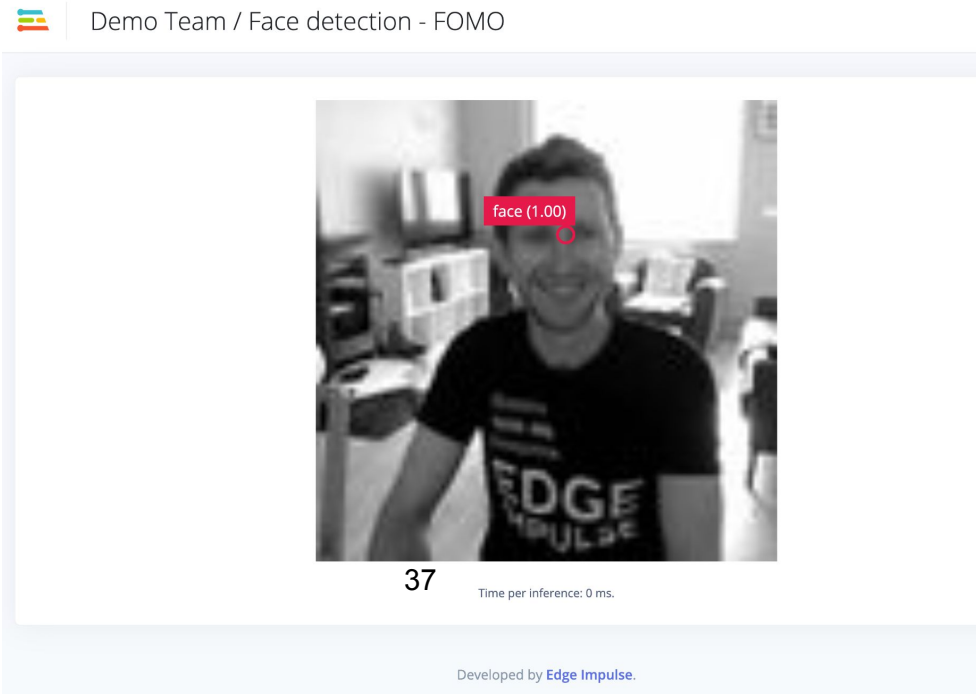


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Object Detection using Centroids

The question the model is trying to answer is:

“Are there faces in the image, **where are they?”**



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Re-train an Object Detection ML model

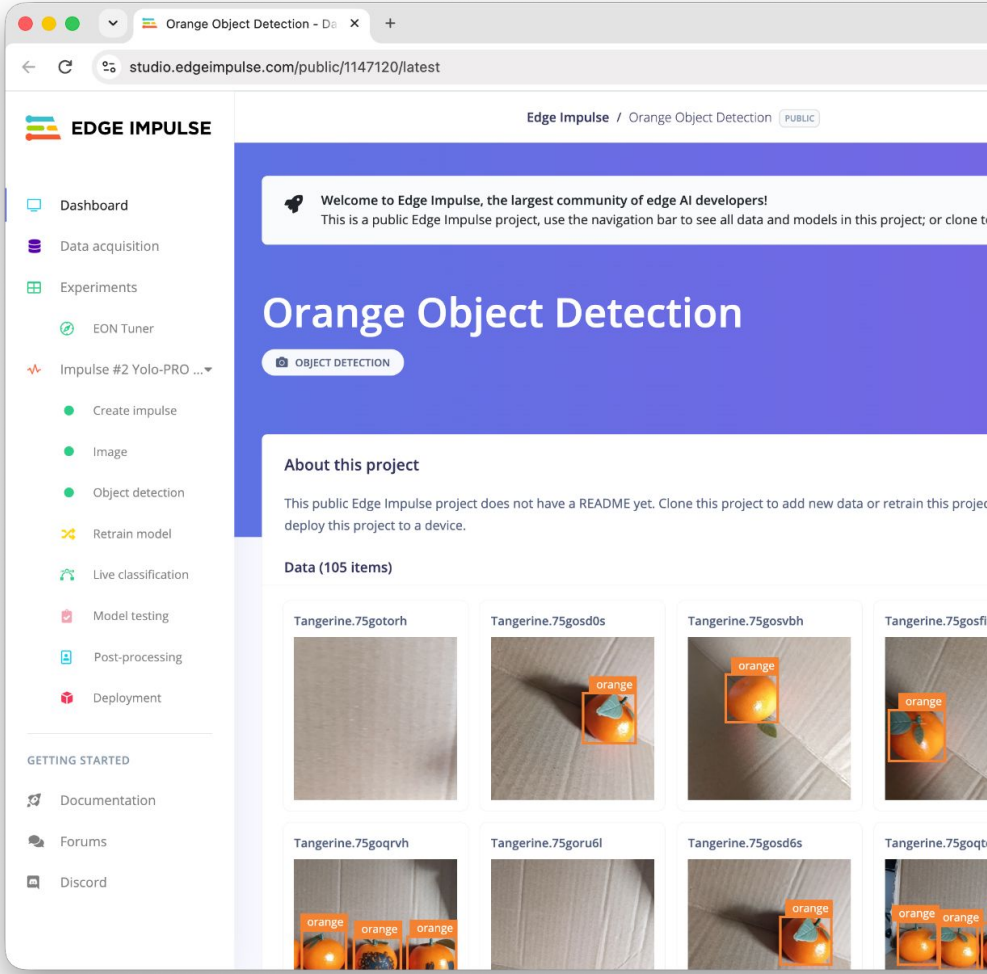
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EDGE IMPULSE

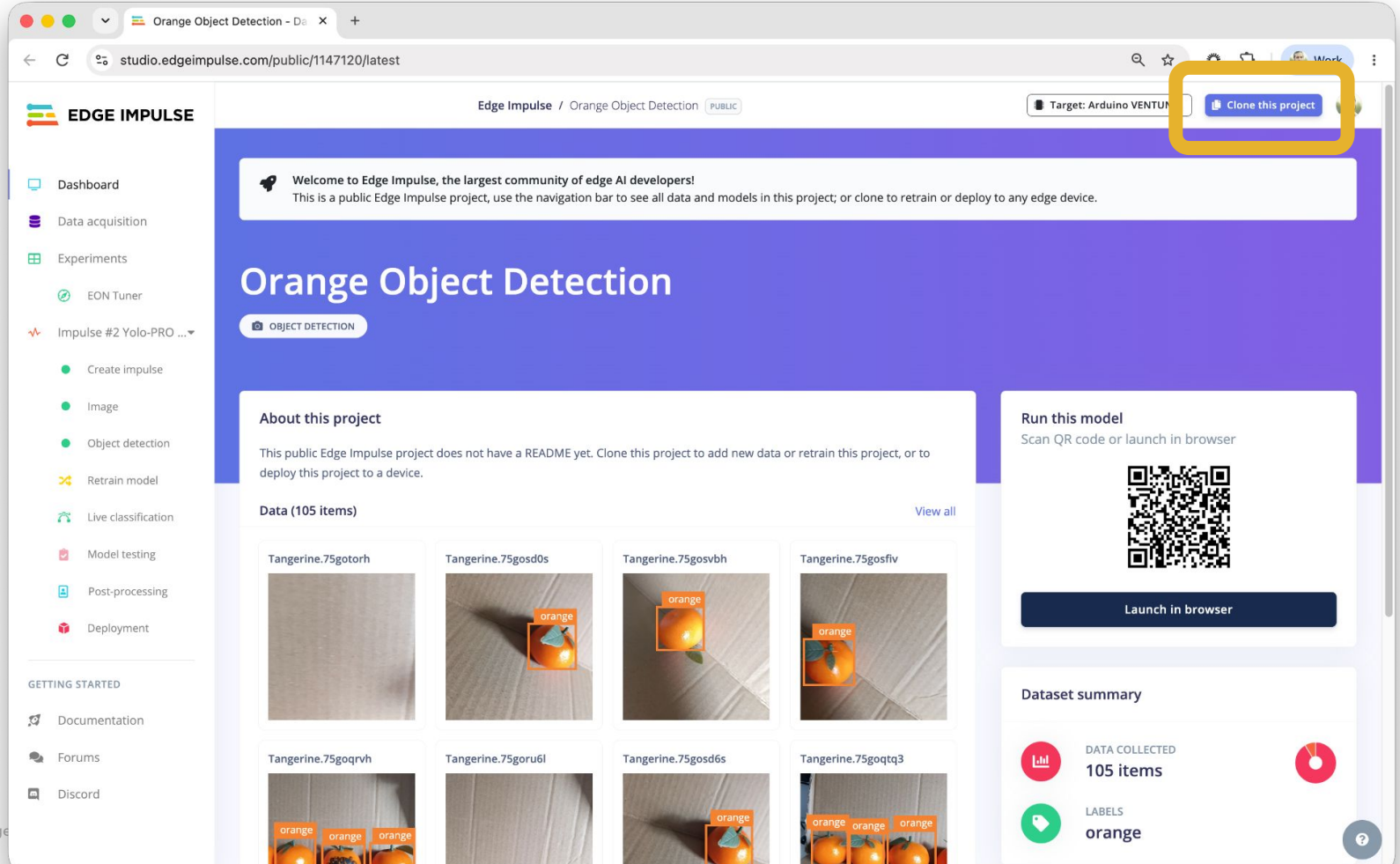
Orange Object Detection

1. Get your laptop
2. Sign-up here

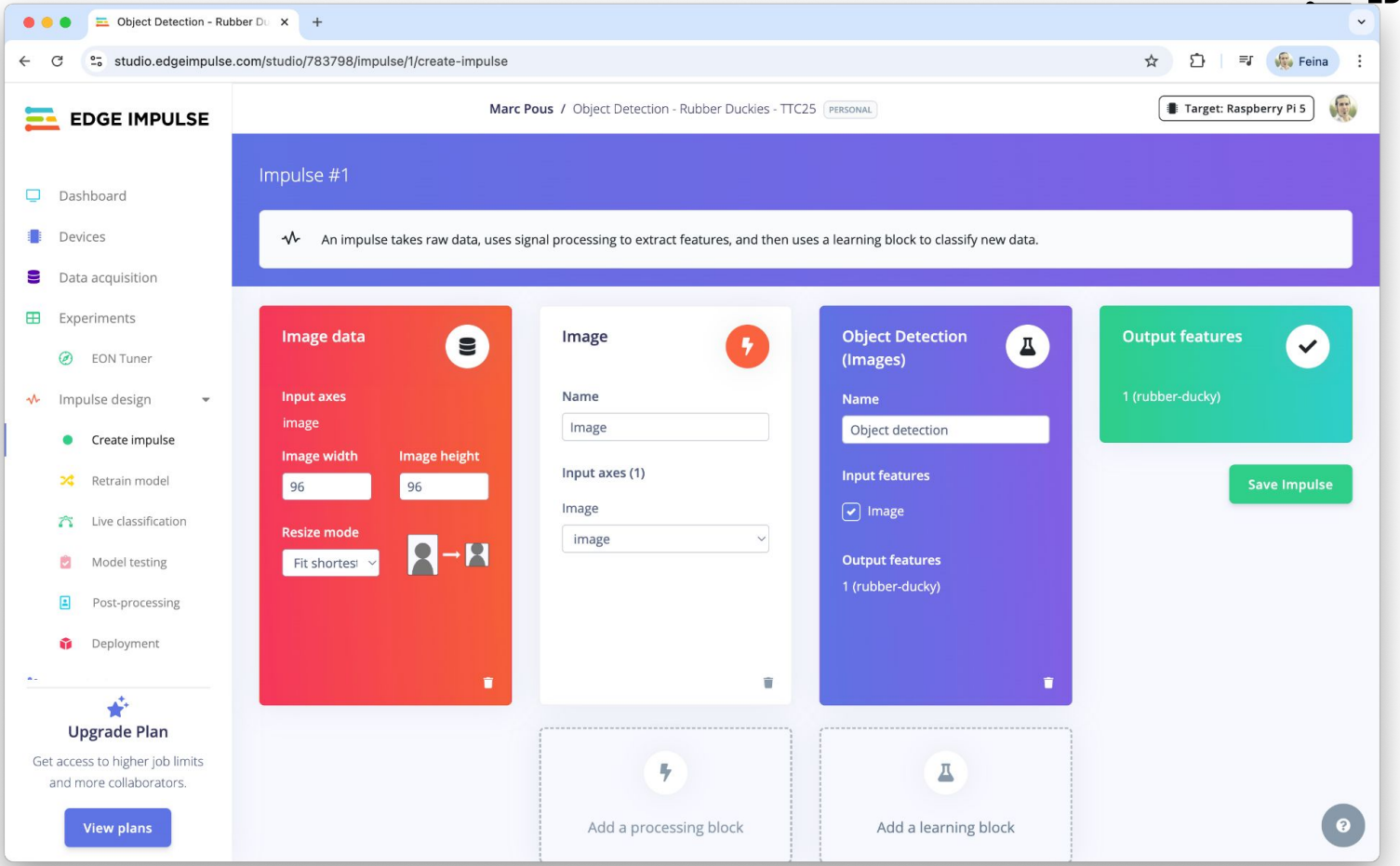
https://edgeimp.com/oranges-roscon26



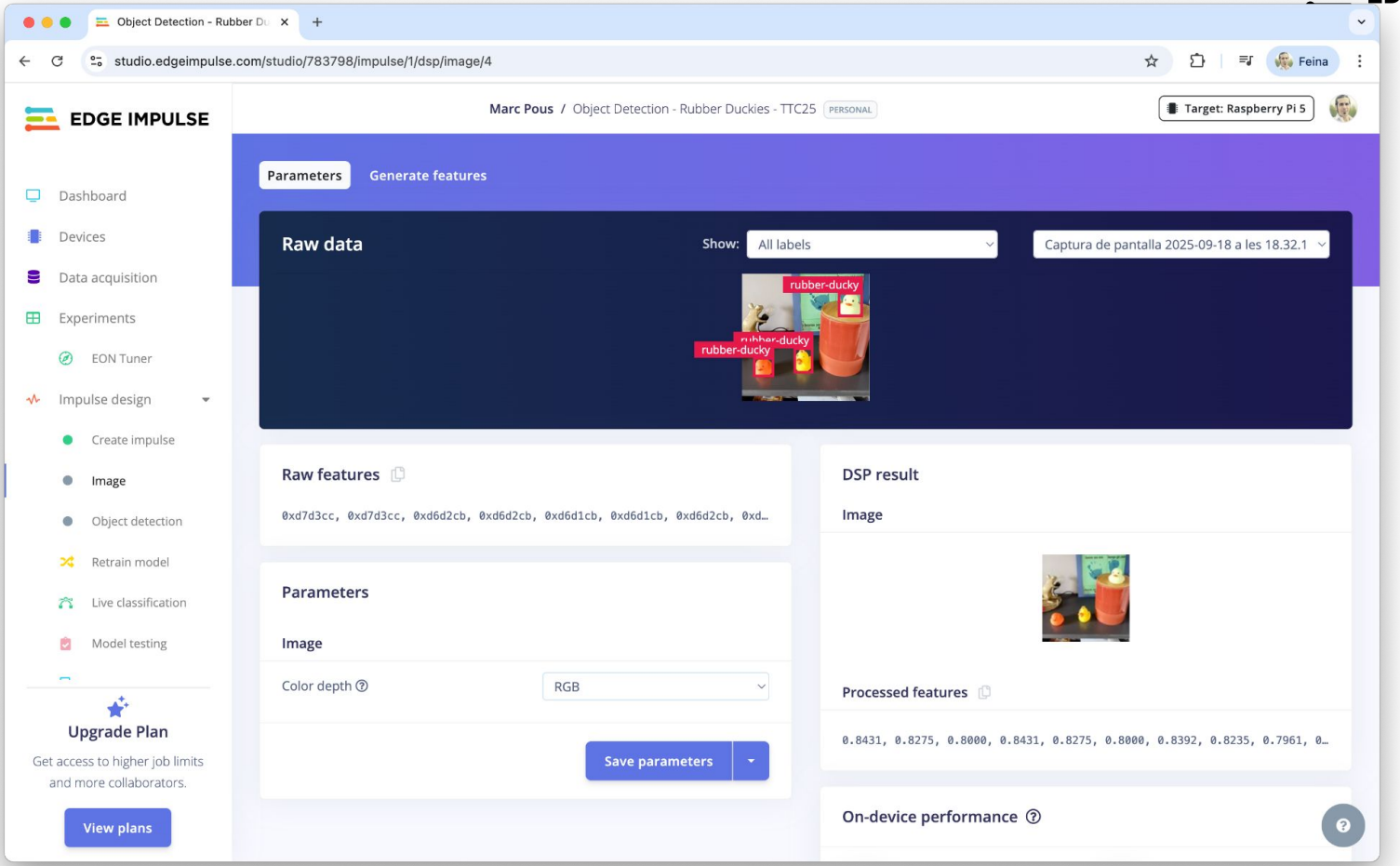
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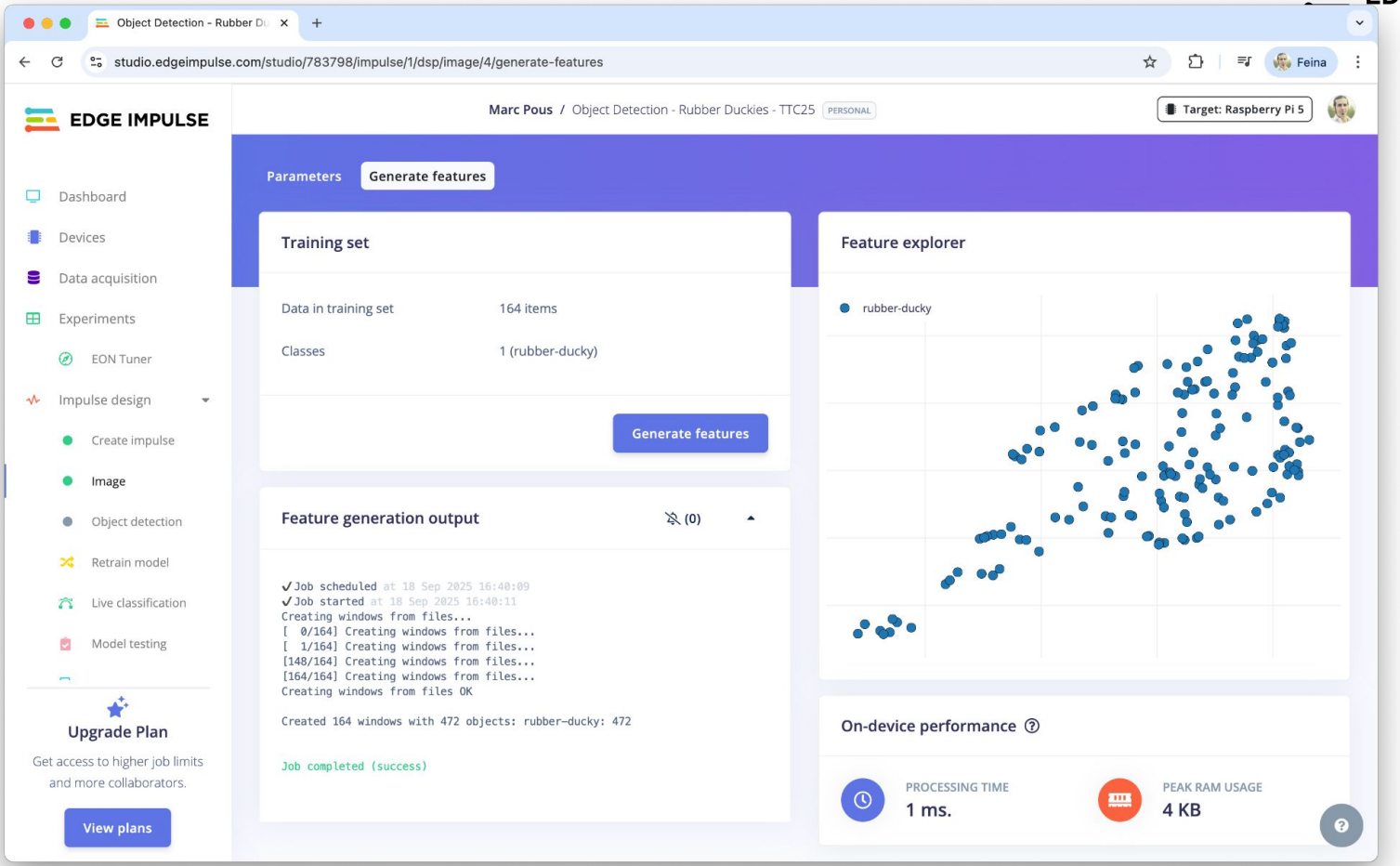
Copyright © Edge



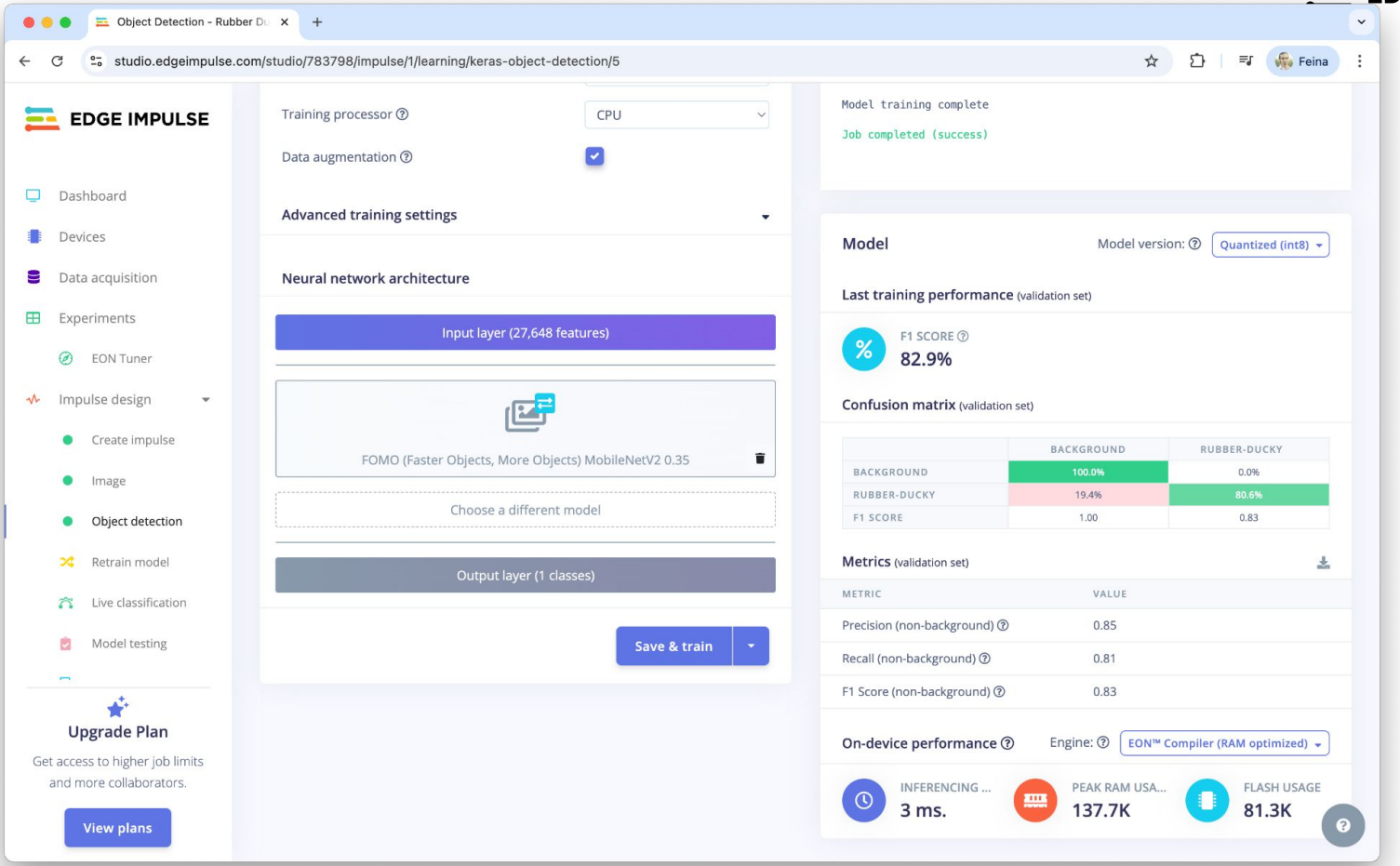
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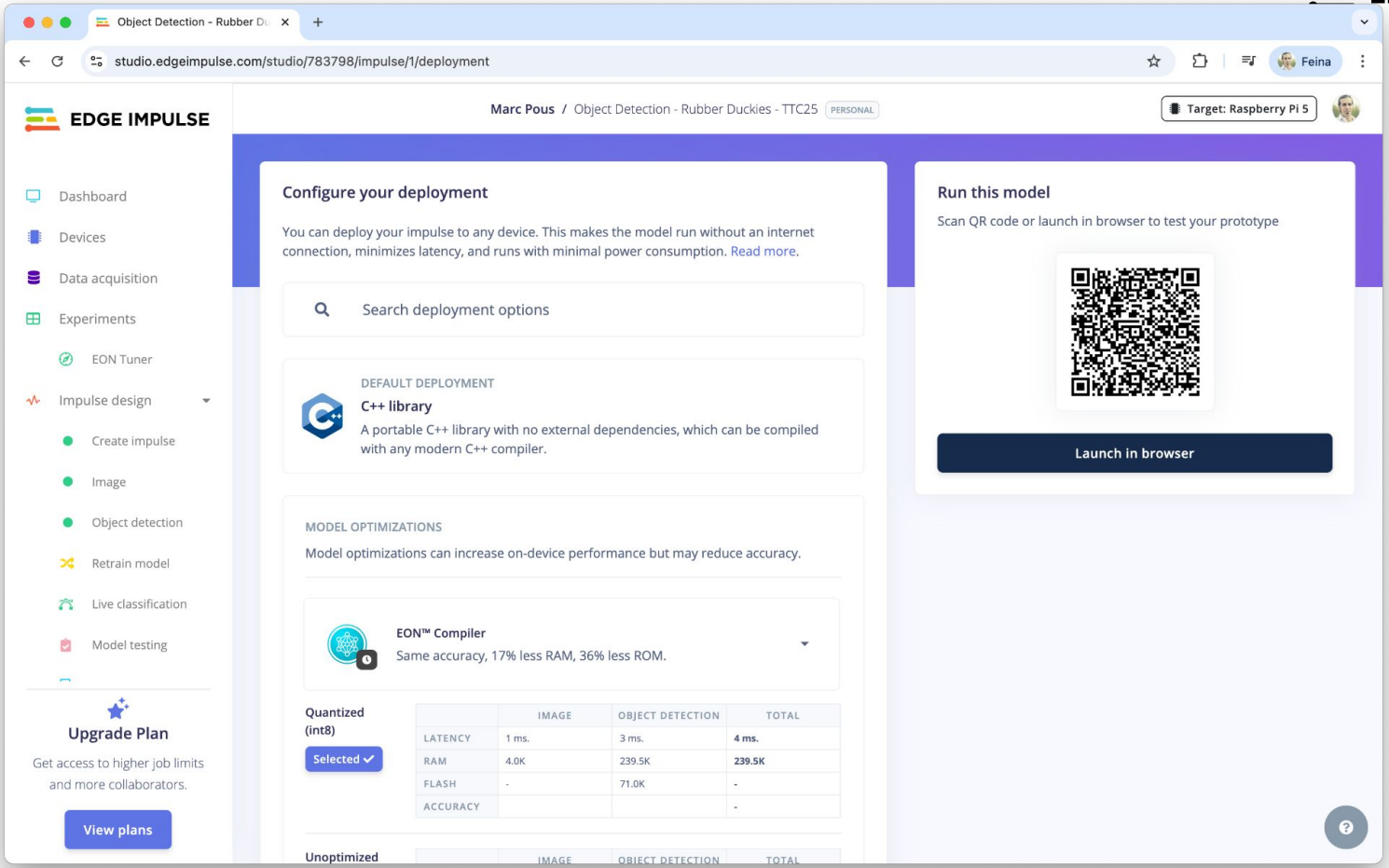
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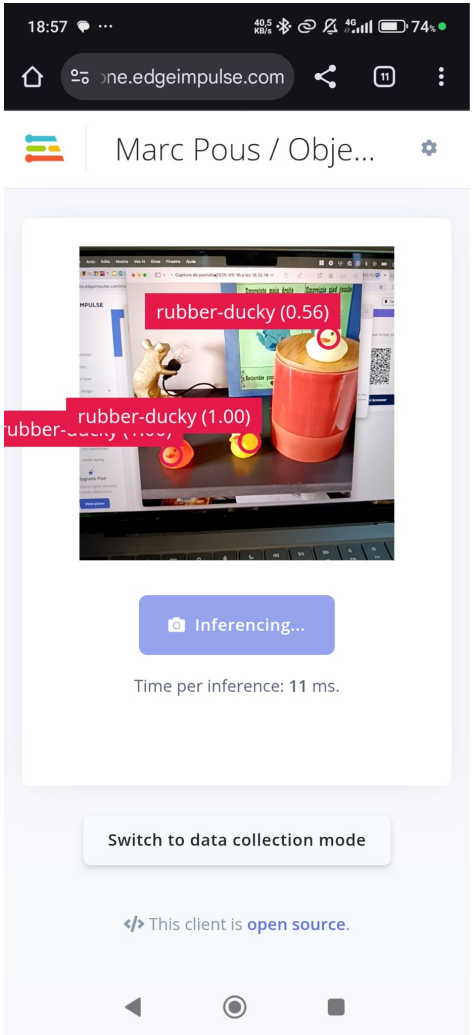
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EDGE IMPULSE

Copyright © Edge Impulse Inc.

but .....




ROSCon ES
ARDUINO
VALENCIA
2026

ROS 2 on the Arduino VENTUNO Q

ROS 2 → Edge Impulse ROS2 node

https://edgeimp.com/gist-roscon-spain



What is Naranjito?





Giovanni Bruno

@br1johnny

github.com/gbr1/

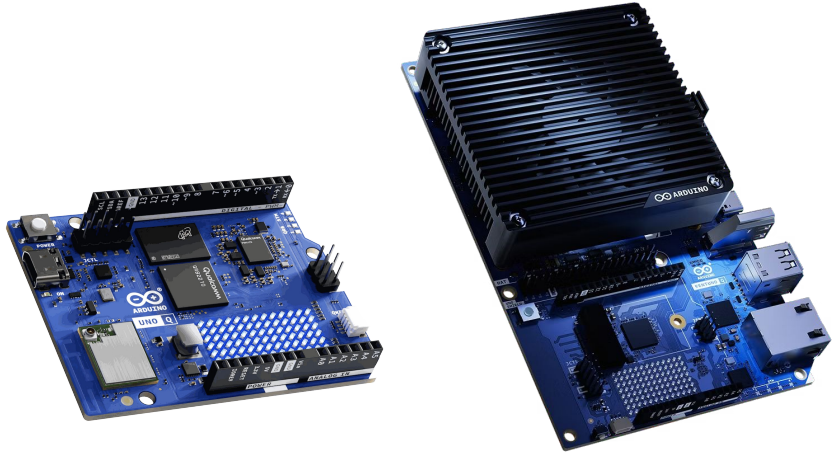


ARDUINO

Arduino RouterBridge ROS:

enabling dual brains boards to ROS 2

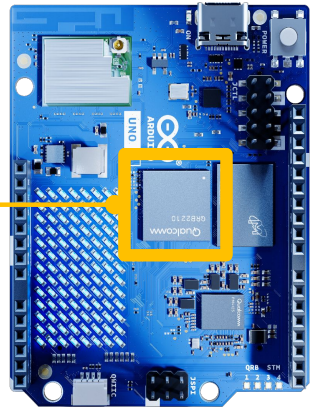
https://github.com/arduino/Arduino_Router_ROS
https://github.com/arduino-libraries/Arduino_RouterBridge_ROS

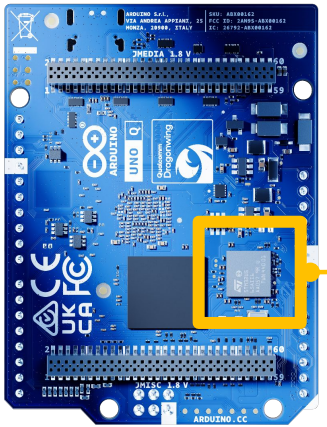


Example: Arduino_RouterBridge

Bridge.provide(“set_status”, set_pin_value)

Qualcomm® Dragonwing ™
Linux: Python





STM32
Zephyr:
Arduino sketch

Bridge.call(“set_status”, led_pin, True)

Arduino_RouterBridge_ROS

call(new_node, publisher, “imu_pub”)
call(new_publisher, sensor_msgs::msg::imu,“/imu/data”)
call(publish_data, “/imu/data”, &data)



Example: Arduino_RouterBridge_ROS

```
#include <Arduino_RouterBridge_ROS.h>

void chatterCallback(std_msgs::msg::String* msg, void* arg) {
    Monitor.print("I heard: [");
    Monitor.print(msg->data.c_str());
    Monitor.println("]");

    (void)arg;
}

class StringSub : public ros2ino::Node {
public:
    StringSub()
    : Node("str_sub_node")
    {
        ros2ino::Subscriber<std_msgs::msg::String>* subscription_ = this->create_subscription<std_msgs::msg::String>
("/chatter",(ros2ino::CallBack)chatterCallback,
            nullptr
            );

        if (subscription_ != nullptr) {
            Monitor.println("Successfully subscribed!");
        } else {
            Monitor.println("Failed to subscribe (List full or Bridge error).");
        }
    }
};

void setup() {
    Bridge.begin();
    Monitor.begin(115200);

    if (!ros2ino::available()) {Monitor.println("No ROS2 proxy available"); delay(1000);}
}

void loop() {
    static StringSub node;
    ros2ino::spin(&node);

    static unsigned long last_serial_print = 0;
    if (millis() - last_serial_print > 5000)
    {
        Monitor.print("string_sub running");
        last_serial_print = millis();
    }
}
```

Support



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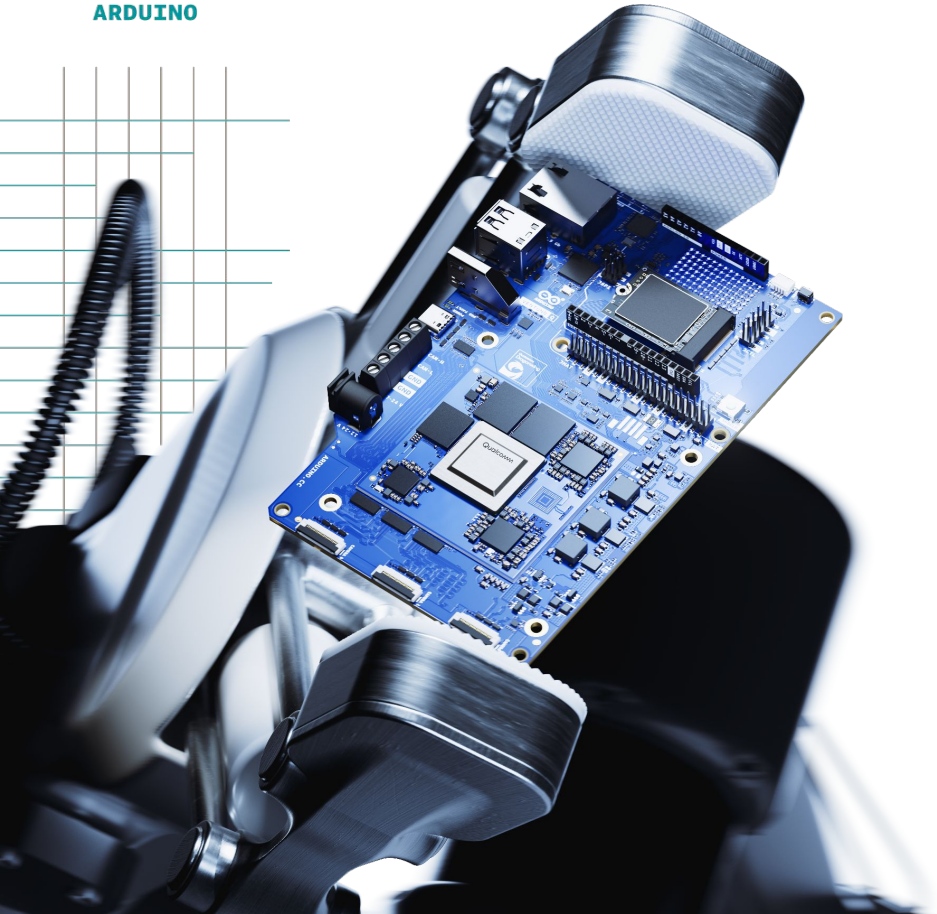
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Product Name	**Arduino® VENTUNO™ Q**
Brand	Arduino
RAM	16 GB
STORAGE	64 GB
SKU	ABX00181
BARCODE	7630049205949
MSRP	**Introductory price $299/€245.08 (excl. VAT)**
General Availability	Aug 25 2026